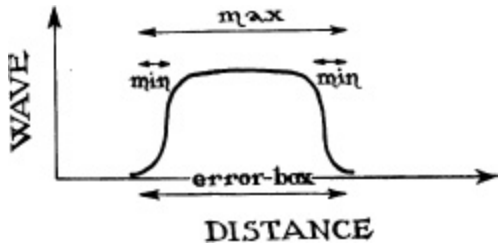
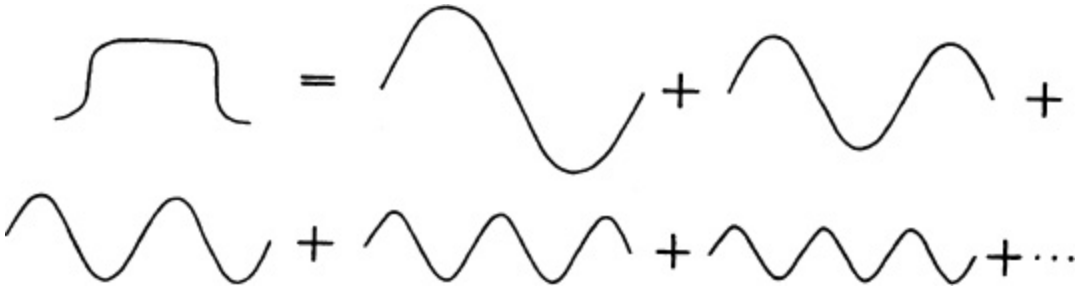
before the measurement, during the measurement the measuring apparatus will “kick” the wave and thereby confine it inside the error box’s interior. The wave, thereby, will acquire a confined form something like the following:



Such a confined wave contains many different wavelengths, ranging from the size of the box itself (marked *max* above) to the tiny size of the corners at which the wave begins and ends (marked *min).* More specifically, the confined wave can be constructed by adding together, that is, superimposing, the following oscillatory waves, which have wavelengths ranging from *max* down to *min:*



Now, recall that the shorter the wavelength of the wave’s oscillations, the larger the energy of the particle, and thus also the larger the particle’s velocity. Since the measurement has given the wave a range of wavelengths, the particle’s energy and velocity might now be anywhere in the corresponding ranges; in other words, its energy and velocity are uncertain.

To recapitulate, the measurement confined the particle’s wave to the error box (first diagram above); this made the wave consist of a range of wavelengths (second diagram); that range of wavelengths corresponds to a range of energy and velocity; and the velocity is therefore uncertain. No matter how hard you try, you cannot avoid producing this velocity uncertainty when you measure the particle’s position. Moreover, when this chain of reasoning is examined in greater depth, it predicts that the more